Not Multi-Agent Only Game genre, evaluation

相关lit 可用.

A Survey on Large Language Model-Based Game Agents

Sihao Hu[†], **Tiansheng Huang**[†], **Fatih İlhan**[†], **Selim Tekin**[†],
Gaowen Liu[‡], **Ramana Kompella**[‡], **Ling Liu**[†]
[†]Georgia Institute of Technology,
[‡]Cisco Research
sihaohu@gatech.edu, ling.liu@gatech.cc.edu

arXiv:2404.02039v1 [cs.AI] 2 Apr 2024

Abstract

The development of game agents holds a critical role in advancing towards Artificial General Intelligence (AGI). The progress of LLMs and their multimodal counterparts (MLLMs) offers an unprecedented opportunity to evolve and empower game agents with human-like decision-making capabilities in complex computer game environments. This paper provides a comprehensive overview of LLM-based game agents from a holistic viewpoint. First, we introduce the conceptual architecture of LLM-based game agents, centered around six essential functional components: perception, memory, thinking, role-playing, action, and learning. Second, we survey existing representative LLM-based game agents documented in the literature with respect to methodologies and adaptation agility across six genres of games, including adventure, communication, competition, cooperation, simulation, and crafting & exploration games. Finally, we present an outlook of future research and development directions in this burgeoning field. A curated list of relevant papers is maintained and made accessible at: https://github.com/git-disl/awesome-LLM-game-agent-papers.

1 Introduction

Intelligence emerges in the interaction of an agent with an environment and as a result of sensorimotor activity.

— *The Embodied Cognition Hypothesis [1]*

Large language models (LLMs), exemplified by ChatGPT [2], represent an important milestone in natural language understanding (NLU) and generative artificial intelligence (Gen-AI). Empowered by generative training over massive data of diverse web sources with hundreds of billions of parameters, LLMs demonstrate astonishing capabilities of generalizing knowledge from huge text corpus data and displaying conversational intelligence in natural language with human-level NLU performance. The emergence of multimodal LLMs (MLLMs), such as GPT-4V [3] and Gemini [4], marks another milestone, enabling LLMs to perceive and understand visual input. We conjecture that the success of LLM technologies fuels an unprecedented opportunity in the pursuit of human-like Artificial General Intelligence (AGI): the cognitive capabilities previously thought to be exclusive to humans, such as reasoning, planning, and reflection, with a degree of self-control, self-understanding, and self-improving, are now achievable by integrating appropriately prompting of LLMs with built-in cognitive intelligence.

We define an LLM-based agent (LLMA) as an intelligent entity that employs LLMs[1] as a core component to conduct human-like decision-making process [5]. Even though LLMAs are capable of cognitive processing similar to human, a distinction between existing LLMAs and human-like

[1]In this paper, LLMs refers to both large language models (LLMs) and multimodal large language models (MLLMs).

AGI is evident: current LLMAs rely on decoding and generalizing pre-existing knowledge derived from pre-training data [6], while AGI is capable of discovering and learning new knowledge through experimentation and experience in real world [7; 8]. Inspired by the process of intelligence development in human infants, the embodied cognition hypothesis [1] posits that the intelligence of an agent emerges from observing and interacting its environment, *i.e.*, grounding the agent in a world that integrates physical, social, and linguistic experiences is vital for fostering conditions conducive to the development of human-like intelligence.

Digital games are recognized as ideal environments for cultivating AI agents due to their complexity, diversity, controllability, safety and reproducibility. Games, ranging from classical chess and poker games [9; 10; 11] to modern video games like Atari games [12], StarCraft II [13], Minecraft [14] and DOTA II [15], have been long instrumental in advancing AI research. Unlike traditional Reinforcement Learning (RL)-based agents [10; 16; 17; 18] that make decisions with the goal of maximizing expected rewards through behavior-level policy learning, constructing LLM-based game agents (LLMGAs) capable of employing cognitive abilities to gain fundamental insights into gameplay, potentially aligns more closely with the pursuit of AGI.

Previous survey papers on LLMs [19; 20; 21] or LLMAs [22; 23; 24] mainly focus on reviewing existing LLMs developed in industry and academic research teams, as well as the general applications of LLMAs, paying less attention to the field of game agents. Concurrent survey papers [25; 26] place a notable emphasis on the game development and cover a limited number of publications on LLMGAs. To bridge this gap, this paper attempts to conduct a comprehensive and systematic survey on recent developments in LLMGAs. Specifically, this survey is organized into three synergistic parts: First, we provide a unified reference framework, in which we describe the essential modules for constructing LLMGAs, covering six core functional components: perception, memory, thinking, role-playing, action and learning. Second, we introduce a taxonomy that categorizes existing literature into six game categories, including adventure, competition, cooperation, simulation, and crafting & exploration. For each category, we describe the technical challenges, the supporting game environments, as well as the commonly used optimization strategies. In the third and final part, we envision different directions of future advancement of LLMGAs.

In summary, this survey paper serves as a comprehensive review of the literature on LLMGAs, offering a taxonomy of six game categories to enhance understanding and facilitate the development and assessment of various LLMGAs. It aims to catalyze progress within this nascent research area and to inspire further innovation in research and development of LLMGAs. Given that this is a new and burgeoning research field, this survey paper will be continuously updated to keep track of the latest studies. A curated list of relevant literature is maintained and accessible at `https://github.com/git-disl/awesome-LLM-game-agent-papers`.

2 A Unified Architecture for LLMGAs

Figure 1 provides an conceptual architecture of LLMGAs that consist of the six essential functional components and their workflow: For each game step, the **perception** module captures game state information, providing the necessary data for the agent to understand its current environment. The **thinking** module processes the perceived information, generating thoughts based on reasoning, planning, and reflection for informed decision-making. **Memory** serves as an external storage, where past experiences, knowledge and curated skills are retained and can be retrieved for future use. The **role-playing** module enables the agent to simulate specific roles within the game, exhibiting believable behaviors that align with each role's characteristics and objectives. The **action** module translates the generated text decisions into executable actions, allowing the agent to interact and manipulate game elements effectively. The **learning** module continuously improves the agent's cognitive and game-playing abilities through accumulated experience and interaction within the game environments.

2.1 Perception

Perception acts like the agent's sense organs, such as eyes, with their primary role being to perceive input from a multimodal domain that encompasses various modalities, including text, visuals, sound, touch, *etc*. Efficient and robust perception functions are critical to empower a game agent to accurately capture the important game state information for decision-making.

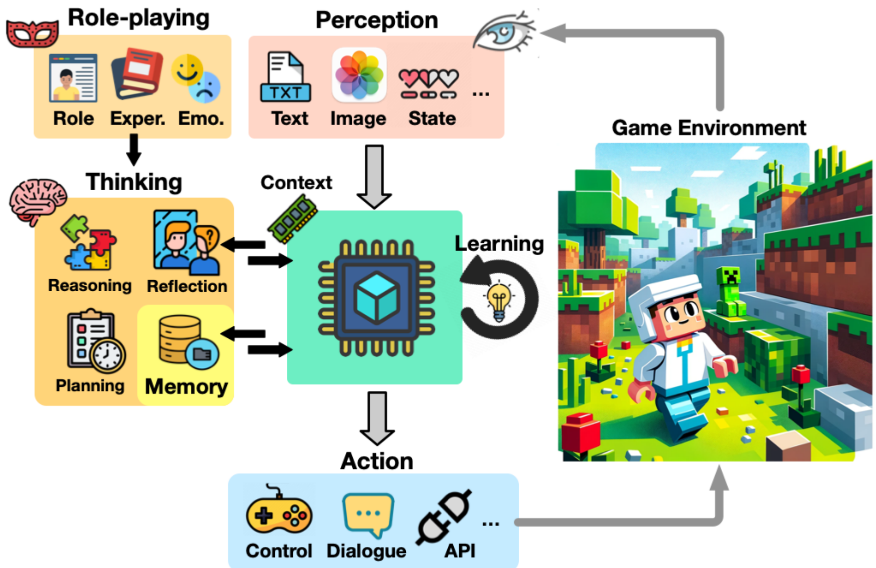


Figure 1: The conceptual architecture of LLMGAs. At each game step, the **perception** module perceives the multimodal information from the game environment, including textual, images, symbolic states, and so on. The agent retrieves essential memories from the **memory** module and take them along with perceived information as input for **thinking** (reasoning, planning, and reflection), enabling itself to formulate strategies and make informed decisions. The **role-playing** module affects the decision-making process to ensure that the agent's behavior aligns with its designated character. Then the **action** module translates generated action descriptions into executable and admissible actions for altering game states at the next game step. Finally, the **learning** module serves to continuously improve the agent's cognitive and game-playing abilities through accumulated gameplay experience.

All text-based games, regardless whether they are text adventure games, such as Zork I [27], or communication games, such as Werewolf [28], are described using natural languages and can be directly tackled by LLMs. In contrast, for videos games there are three primary ways to enable agents perceive the game state:

1. **State variable access**: Some game environments [29; 30; 31; 14; 32] support to access symbolic state variables via internal APIs. For example, a Pokémon in Pokémon battles [30] can be represented by the state variables of species, statistics, status, and moves, without relying on any visual information. In Minecraft, Mineflayer [33] provides high-level APIs to access the local environment state, such as positions, blocks, inventory. The state values are filled into designed prompt templates to form textual descriptions of game states. However, not all the games support internal APIs, and describing games merely with symbolic states can result in information loss, especially for games that require detailed visual information to fully capture the gameplay experience, like Red Dead Redemption 2 [34] and StarCraft II [29].
2. **External visual encoder**: To solve text-only problem, existing studies equips LLMs with external visual encoders to translate visual information into textual observations. For example, CoELA [35] and LLMPlanner [36] adopt object detectors to recognize objects within the agent's field of view in embodied environments. The CLIP [37] visual encoder and its variants are widely used for mapping images into pre-defined text descriptions [38; 39; 40]. For example, JARVIS-1 [39] uses MineCLIP [41] to select the most similar text description from a set of 1,000 Minecraft text data entries for images; ELLM [40] adopts ClipCap [42] as the captioner for visual observations: it maps CLIP embedding to a 10-token sequence, which are fed as the prefix for GPT-2 to generate the whole caption.
3. **Multimodal LLMs (MLLMs)**: Visual encoders fall short in generalizability for unseen scenarios or objects, as they primarily rely on predefined text descriptions for classification. In comparison, MLLMs align visual and textual information in a unified representation space and decode them into natural languages, thereby enabling better generalizability across un-

known scenarios. General-purpose MLLMs like GPT-4V [3] are adopted in the game-playing of RDR2 [34], Doom [43], Minecraft [44] and simulated Embodied household [45] to directly perceive visual observations for decision-making or generating text data as the perception module, but usually need error correction mechanisms [45; 34] with feedback from the environments to address inaccuracies; Game-specific MLLMs involve supervised learning on multimodal instruction data generated by experts, such as GATO [46] and SteveEye [47], or learned from environmental feedback through RL such as Octopus [45].

In summary, for video games, accessing symbolic states requires the support of internal APIs. External visual encoders suffer from limited generalizability, as they cannot fully cover all scenarios or objects, especially those without predefined textual descriptions. Although general-purpose MLLMs address above-mentioned issues, they are still insufficient for distinguish fine-grained details like the relative positions of target objects, and struggles to understand game-specific concepts [34]. Grounding and disciplining MLLMs with game experience and feedback [45] is a promising way to enable better perception and understanding for games.

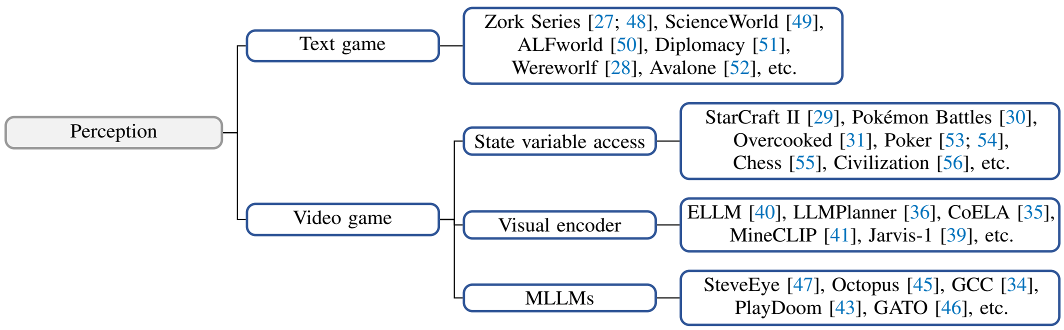


Figure 2: Mind map for the perception module.

2.2 Memory

Humans rely on memory systems to memorize prior experiences for recalling, learning, and applying knowledge in future scenarios. Similarly, LLMGAs necessitate memory systems to ensure operational coherence and efficiency, serving as a repository for their past observations, thoughts, actions, and skills, from which agents retrieve essential information for strategy formulation and decision-making.

From a perspective of cognitive science [57; 5], human's memory mechanism can be divided as working memory and long-term memory, where working memory stores an agent's current context, and long-term memory stores the agents past experience and thoughts. For LLMGAs, working memory can be regarded as the context of LLMs, and the term "memory" here refers to the long-term memory, which acts as an external storage. Long-term memory stores **episodic** memories [58] such as observation streams [59] and previous game trajectories [28; 60; 61] generated through the perception module; high-level **semantic** memories [62] that represents the agents' understanding of itself and the game world [59; 63], generated through the thinking module; and **procedural** memories [64], which represent curated skill stored as code [65; 34] or plans [66; 39].

Retrieval: As memories accumulate over time, the majority of them distract from decision-making. Retrieval serves as an essential role to filter out through and pass the most relevant memories to the agent. Memory records are typically stored as key-value pairs. In semantic retrieval, the process involves calculating the semantic similarity between the representations of a query and the memory keys, and selecting the memory values with the highest similarity to the query object. The query object can be various forms, such as self-instructed questions [59], task-triggered questions [65], predefined questions [28], or visual observations [39]. Specifically, in Voyager [65]'s memory system, the keys are program descriptions, and the values are the previously executed successful program codes. In JARVIS-1 [39], the keys are composed of task descriptions paired with observations in images, while the values represent the previously executed plans. Additionally, to simulate the human forgetting mechanism, Generative Agents [59] take into account recency and importance, where recency is calculated using an exponential decay function over game hours, and importance is evaluated by the LLM to differentiate mundane details from core information.

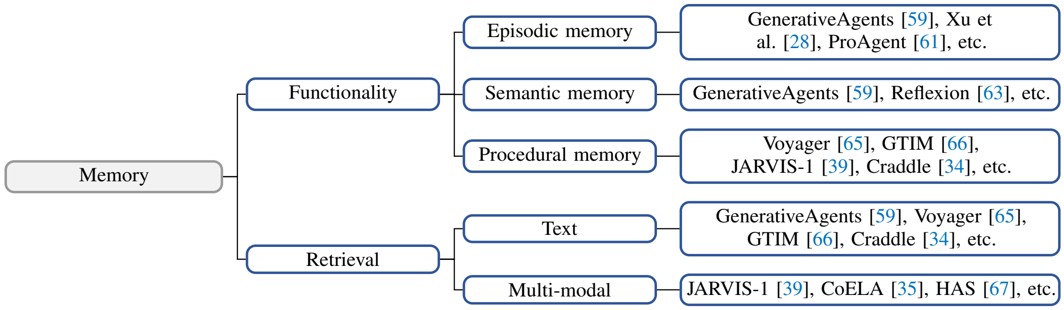


Figure 3: Mind map for the memory module.

2.3 Role-playing

Role-playing enables agents to assume diverse characters or roles within the game, generating believable conversations and behaviors appropriate to the given roles. Many games feature role-playing elements [59; 28; 52] where players assume specific roles and engage in game-playing from the perspective of the characters, leading to immersive gaming experiences. Role-playing is also important for building Non-Player Characters (NPCs) [68] and game assistants [69], as well as for generating dialogues [70].

It has been proved that assigning different personality types can largely influence the generative style of LLMs [71; 70]. Role-playing can enhance the vividness [72], personalization [73] and proficiency [74] of LLMAs, and generating dialogues with affective information makes agents' behavior more believable [75; 76]. For role-playing, the simplest way is to directly insert natural language descriptions of a role's identity, such as character traits, hobbies, occupation and social relationships, as initial memories for the agent [59]. Evaluations show that providing few-shot dialogue examples or fine-tuning can further enhance the role-playing performance in conversational tasks [70; 77]. Recent advanced approaches such as CharacterLLM [78] build imaginary experience from characters' profiles, and fine-tune LLMs with these experiences to enable agents to exhibit consistent personalities and express emotions.

2.4 Thinking

Thinking is the cognitive process of analyzing and integrating information. In this section, we introduce two primary thinking methods for decision-making: reasoning and planning. Reasoning involves using deduction, induction, and abduction to generalize observations, derive conclusions, and infer explanations. In comparison, planning strategizes decision steps to achieve complex and long-horizon goals.

2.4.1 Reasoning

Reasoning [79] is a process that starts from observation, factual evidence, and previous thoughts, then progresses through analyzing and synthesizing these elements to deduce specific conclusions from general principles (deduction), infer general principles from specific instances (induction), or formulate the most likely explanations (abduction). Reasoning is fundamental to human cognition, enabling individuals to make sense of the world, solve problems, and make informed decisions.

LLMGAs [80; 56; 43; 30] adopt general-purpose reasoning approaches [81; 82; 80; 83; 84] to analyze information logically, providing informative insights for decision-making: ReAct [80] introduce reasoning to condition the generation of action with few-shot prompting; CoT [81] and Zero CoT [82] decompose the entire thinking process into multiple chained thoughts, enabling the step-by-step elucidation of complex problems; SC [84], ToT [83] and GoT [85] follow a multi-path reasoning paradigm: SC conducts multiple times independent reasoning and choose the result with the highest frequency as the final output; ToT [83] and GoT [85] organize reasoning paths into tree and graph-like structures to enhance the reasoning ability.

Reflection [63; 38; 86] can be recognized as a special type of reasoning that usually occurs after feedback from previous trials is provided. It involves the agent analyzing its own actions, decisions, and thought processes, and considering how these could be improved based on the feedback received. This reflective process allows for the adjustment and refinement of strategies and behaviors, contributing to learning and development over time. Specifically, Reflexion [63], DEPS [38], AgentPro [61], ProAgent [87] identify errors and inefficiencies in past failed attempts through self-reflection and reuse the thoughts to enhance the performance in subsequent trials. Moreover, RCI [88], Self-Refine [86] and GPTLens [89] demonstrates that the feedback can not only come from environments, but also LLMs themselves, and iteratively refine the results by incorporating self-generated reflection.

In game playing, Hu et al. [30] discover that when encountering a powerful opponent, CoT can introduce panic feelings, causing the agent to act inconsistently, such as switching to different Pokémon in consecutive turns. In comparison, SC alleviates the issue by voting for the most consistent action; Theory-of-Mind [90; 91] (ToM) thinking involves inferring others' intentions from a shifted perspective, and demonstrate enhancement in imperfect information games like Poker [54] and enables LLMGAs recognize partners' intention for assistance in cooperation games [92]. Moreover, reflecting on the surface observations or experiences can provide high-level, abstract thoughts, which helps the agent act more reasonably and believably [59].

2.4.2 Planning

Humans utilize planning as a strategic tool to address and manage challenging and long-term tasks. For LLMGAs, planning involves the decomposition of a complex task into simpler, executable subtask set. Existing planning approaches can be categorized as goal-free planning and goal-conditioned planning, based on whether a predefined goal is necessary for the planning process.

Goal-free planning: Open-ended games usually do not have prefixed goals for players to achieve. Generating a goal plan saves the agent from being overwhelmed by numerous possible actions. Existing goal-free planning approaches [59; 93; 65; 40; 94; 95] primarily instruct LLMs to generate goal plans. Voyager [65], ELLM [40], SPRING [94] and AdaRefiner [95] prompt LLMs with agent's states such as hunger, inventory, and equipment, and local observations for generating suitable next goals; OMNI [93] prompts LLMs to select interesting and learnable tasks for agent to explore the open-world. In simulation games, a long-term daily plan can effectively prevent incoherent behaviors [59; 30]. Generative Agents [59] utilize a top-down approach for generating a one-day plan for human-simulacra agents, starting with a broad initial plan for the day, then breaking it down into more detailed action plans. After planning, agents can choose to either continue with the plan or react to its dynamic local environment.

Goal-conditioned planning: A goal-conditioned planner recursively translates a goal, task, or instruction into a set of subgoals until it reaches structured actions. Goal-conditioned planning is used for addressing long-horizon and complex tasks such as crafting tools [38; 66] or performing quests [36; 96; 97]. Existing studies primarily instruct LLMs to generate plans. ZeroShotPlanner [98] and LLMPlanner [36] prompts LLMs with zero-shot or few-shot examples for planning; Given the difficulty in generating a correct plan on the first attempt, GITM [66] and JARVIS-1 [39] leverage external knowledge, such as item crafting recipes to enhance planning, and also incorporate environmental feedback such as error messages to refine the initial plan; DEPS [38] introduce error correction on initial plans by integrating description of the plan execution and self-explanation of feedback when encountering failures; Adapt [96] and SwiftSAGE [97] adaptively decompose tasks with LLMs when encountering execution failures; S-Agents [99], HAS [67] and MindAgents [100] operate in a hierarchical cooperation structure in which a LLM planner dispatches sub-tasks to multi-agents for efficient execution.

2.5 Action

After humans make decisions to take actions, they control their bodies, such as hands, to execute these actions, translating cognitive decisions into physical movements that interact with the world around them. The action module serves as the hands of the LLMGAs, translating language-described decisions into executable actions in the game environment, enabling the agents to interact with and alter their surroundings or game states. Different games necessitate different levels of granularity in agents' output actions. For games requiring manipulative control like RDR2 [34], Minecraft [14]

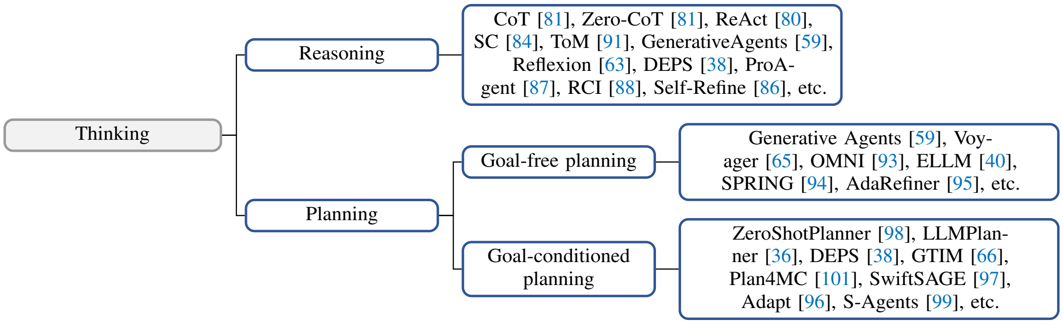


Figure 4: Mind map for the thinking module.

and Overcooked [102], a low-level action like keyboard or mouse operation is required. In contrast, games without manipulative control like text adventure games [103; 104], Pokémon battles [30] and Poker [53] directly facilitate the execution of high-level actions.

LLMs typically generate high-level actions rather than low-level actions. Therefore, for games **with manipulative control**, a translation module is required to translate LLM-generated action into low-level actions. Existing studies adopt heuristics [92; 105; 66; 33; 59] or RL policies [40; 101; 106] for translating a high-level action into low-level action sequences. Heuristic-based translation generates low-level movements using path-finding algorithms, along with manipulative actions. For example, in Overcooked, given a high-level action "chop tomato", the translation module identifies the shortest path to the target with a breadth-first search algorithm and identifies a sequence of movements along with the *chop* action [92; 105]; In Minecraft, the high-level "approach" action uses an A* algorithm for path-finding and executes low-level actions like jump, move and fall in four directions [66; 33]; In comparison, RL-based approaches [41; 40; 101; 106] train language-conditioned RL policies that take observations and high-level actions as input to generate low-level actions, rewarded based on the semantic similarity between the goals and the agent's transitions.

Games **without manipulative control** can be divided as parser-based games [27; 107] and choice-based games [30]. Parser-based games require LLMs to generate an action word by word, wherease choice-based games only need LLMs to select from a set of given actions. For parser-based games, ZeroShotPlanner [98] proposes semantic translation that maps LLM-generated free-form actions to the semantically similar, admissible actions; SayCan [108] calculates the probability of each admissible action using the chain rule by multiplying the conditional generation probability of each successive string given the previous string.

2.6 Learning

Humans are able to refine their cognitive abilities and acquire knowledge by interacting with the physical world, gaining hands-on experience through direct engagement with their environments. Similarly, an LLMGA's learning process involves improving its cognitive and game-playing abilities over time, based on the experiences and feedback received from the game environment.

LLMs encode a wealth of semantic knowledge about the world while lack of real experience within environments, *i.e.*, they are ungrounded [108]. The majority of existing LLMGAs adopt frozen LLMs to play games, relying on carefully designed prompts [28; 109] or external knowledge [30; 39; 66]. In comparison, enable LLMGAs to learn in environments is crucial, since it closely mirrors the way humans acquire knowledge through interacting with the real world. Existing learning approaches can be divided into three categories: in-context feedback learning, supervised fine-tuning and reinforcement learning.

In-context feedback learning: Feedback represents a type of evaluation for previous strategies. By including feedback from environments into context, LLMs are able to iteratively "reinforce" strategy generation without updating weights [63; 65; 66; 30]. Specifically, Reflexion [63] and DEPS [38] generate self-reflection/explanation on the feedback like failure signal and reuses the thought for the next trail; Voyager [65] and GTIM [66] iteratively prompt LLMs to re-generate action code with error messages; Hu et al. [30] uses manually generated feedback such as the HP change across consecutive turns as evaluation for previous actions; Furthermore, existing works [63; 86; 86] demonstrate feedback cannot only comes from the game environments, but also from LLMs themselves.

Supervised fine-tuning: Supervised fine-tuning [110; 111] gathers high quality experience to fine-tune LLMs, based on the assumption that such experiences encompass environmental knowledge. Specifically, E2WM [110] collects embodied experience in VirtualHome with Monte Carlo Tree Search and random exploration; LLAMARider [111] gathers experience in Minecraft via self-reflection with feedback. Both of them demonstrate that fine-tuning on the collected experience enhances capability of LLMs on solving tasks within the environment. Moreover, imitation learning-based approaches like GATO [46], LID [112], SwiftSAGE [97] and Octopus [45] fine-tune LMs using expert or oracle trajectories to enhance their performance as policies.

Reinforcement Learning: Existing RL-based LLMGAs can be divided into three categories: (1) **LLM as actor**: GLAM [113] is grounded in the BabyAI-text environment as a policy to select next action (four movements), training through online RL [114]; (2) **LLM as planner**: Existing studies such as SayCan [108], Plan4MC [101], RL-GPT [106], ELLM [40], follow a hierarchical paradigm that integrates fixed LLMs as high-level planners with separate low-level RL policies to execute actions. In comparison, another line of research involves fine-tuning large language model (LLM) planners based on rewards received from the environment, such as Octopus [45]; (3) **LLM as presenter**: LMs can be co-trained with RL policies to produce consistent dialogues that reflect the intentions of policy models, especially in communication games such as Diplomacy [51] and Werewolf [115]; (4) **LLM for reward design**: LLMs can directly serve as reward models [116], provide annotations for training reward models [117], or generate and refine reward functions for guiding RL agent training [118; 119; 120].

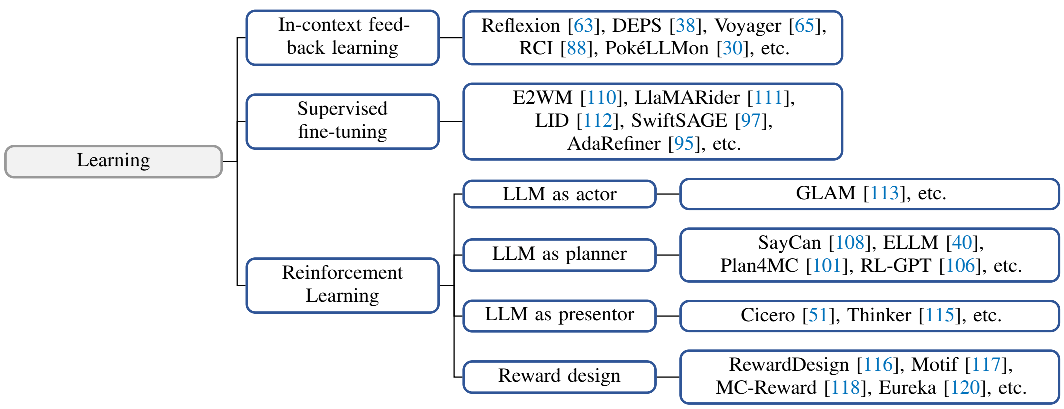


Figure 5: Mind map for the learning module.

3 LLMGAs in Games

We categorize existing studies into six categories based on the main characteristics of the games they support, including adventure, communication, competition, cooperation, simulation, and crafting & exploration. Figure 6 illustrates the core gameplay mechanics associated with its genre:

- **Adventure**: Adventure games emphasize story-driven gameplay, where players explore environments, solve quests and interact with characters and objects to progress the game. Representative games: Zork I [27] and Red Dead Redemption 2 (RDR2) [34].
- **Communication**: Communication games revolve through the turns of communication, negotiation, deduction and even deceptions among multiple players. Representative games: Werewolf [28] and Diplomacy [51].
- **Competition**: Competition games pit players against each other in challenges that test skill or strategy, aiming to outperform others for victory. Representative games: StarCraft II [29] and Pokémon Battles [30].

Figure 6: The depiction of six game categories.

- **Cooperation**: Cooperation games are designed around players working together towards common goals, emphasizing teamwork, collaborative problem-solving, and shared achievements. Representative games: Overcooked [102].
- **Simulation**: Simulation games replicate real-world events in detail, allowing players to experience and manage scenarios ranging from building a civilization or living another life. Representative games: The Sims [59; 121] and Civilization [56].
- **Crafting & Exploration**: Crafting & Exploration games provide open worlds where players gather resources, craft items, and exploring within expansive environments, encouraging creativity and discovery. Representative games: Minecraft [14] and Crafter [122].

We summarize existing studies on LLMGAs in Table 1. In this section, we will walk through six game categories, highlighting key findings and methodologies employed in the current research landscape.

3.1 Adventure Games

Adventure games typically progress through storylines or quests. We categorize existing works into two types based on modality: text-based adventure games and video adventure games.

Text adventure games: A text adventure game provides a text-based environment in which players use text commands to interact with the world, exploring and completing quests. TextWorld [137] is a generator of synthetic text games [138; 103] with varying difficulty levels by adjusting parameters such as the numbers of rooms and objects, quest length and complexity; Jericho [104] is a collection of 56 human-made games originally designed for human players, covering fictions such as the *Zork* series [27; 48] and *Hitchhiker's Guide to the Galaxy* [139]; ALFWorld [50] is aligned to the embodied environment ALFRED [140], where agents are requested to accomplish six types of household tasks; ScienceWorld [49] simulates a primary school science curriculum, such as thermodynamics and electrical circuits. To complete a quest, an agent needs to navigate to specific rooms, obtain necessary items, conduct experiments, and analyze the results; BabyAI-Text [113] is a text extension of BabyAI [141], a procedurally generated minigrid environment where an agent navigates and interacts with objects.

Due to the lack of graphics, text games rely on the commonsense knowledge as a prior for how to interact with the environment. In parser-based text games, generating a three-word sentence with a small vocabulary of size 1,000 leads to 1 billion combinatorial candidates. Pre-trained LMs featuring human knowledge can effectively narrow down the action space and thus have been widely utilized as linguistic priors for guiding RL agents [123; 142; 143; 144]. Recently, LLMGAs are employed to playing text adventure games: Tsai et al.[124] suggest that the game-playing ability of GPT-3.5 is on par with state-of-the-art (SOTA) reinforcement learning (RL) approaches[145; 146], but it is incapable of constructing the entire map of a partially-known environment; REACT [80] and Reflexion [63] prompt LLMs to generate additional reasoning and reflection to condition the generation of actions; To solve challenging tasks, Adapt [96] and SwiftSage [97] adopt an LLM planner to decompose complex tasks into subgoals as needed; GLAM [113] leverage online RL to ground an LLM in BabyAI-text as a policy.

Video adventure game: Red Dead Redemption 2 (RDR2) is a 3D action-adventure game in which players assume the role of an outlaw, and follow the storyline of his life as part of a criminal gang. The game features an important characteristic, i.e., it guides the player what to do next with instant instructions. Cradle [34] is an LLMGA that perceives the game screen, analyzes instructions, generate action plans and controls the character through mouse/keyboard operations using GPT-4V.

Table 1: Comparison among existing LLMGAs. FT denotes Fine-Tuning.

Studies	Category	Game	Base Model	FT	Modality
CALM [123]	Adventure	Jericho	GPT-2	✓	Txt
CanPlayWell [124]	Adventure	Zork I	GPT-3.5	✗	Txt
ReAct [80]	Adventure	ALFWorld	PaLM	✗	Txt
Reflexion [63]	Adventure	ALFWorld	GPT-3	✗	Txt
ADAPT [96]	Adventure	ALFWorld	GPT-3.5	✗	Txt
SwiftSAGE [97]	Adventure	ScienceWorld	GPT-4 & T5	✓	Text
GLAM [113]	Adventure	BabyAI-Text	FLAN-T5	✓	Txt
Cradle [34]	Adventure	RDR2	GPT-4V	✗	Txt & Img
Xu et al. [28]	Communication	Werewolf	GPT-3.5	✗	Txt
Xu et al. [125]	Communication	Werewolf	GPT-4	✗	Txt
Thinker [115]	Communication	Werewolf	ChatGLM-6B	✓	Text
ReCon [52]	Communication	Avalone	GPT-4	✗	Txt
AvaloneBench [126]	Communication	Avalone	GPT-3.5	✗	Txt
CodeAct [127]	Communication	Avalone	GPT-4	✗	Txt
Cicero [51]	Communication	Diplomacy	BART	✓	Txt
WarAgent [109]	Communication	Diplomacy-like	GPT-4	✗	Txt
CosmoAgent [128]	Communication	Diplomacy-like	GPT-4	✗	Txt
DEEP [129]	Communication	Word Guess	GPT-4	✗	Txt
GameEval [130]	Communication	Word Guess	GPT-4	✗	Txt
PokéLLMon [30]	Competition	Pokémon Battles	GPT-4	✗	Txt
CoS [29]	Competition	StarCraft II	GPT-3.5	✗	Txt
SwarmBrain [131]	Competition	StarCraft II	GPT-3.5	✗	Txt
ChessGPT [55]	Competition	Chess	RedPajama-3B	✓	PGN
OthelloGPT [132]	Competition	Othello	GPT	✓	PGN
PokerGPT [53]	Competition	Texas Hold'em	OPT-1.3B	✓	Txt
GoodPoker [133]	Competition	Texas Hold'em	GPT-4	✗	Txt
SuspicionAgent [54]	Competition	Leduc Hold'em	GPT-4	✗	Txt
AgentPro [61]	Competition	Leduc Hold'em	GPT-4	✗	Txt
LLM-Co [92]	Cooperation	Overcooked	GPT-4	✗	Txt
MindAgent [100]	Cooperation	Overcooked	GPT-4	✗	Txt
ProAgent [87]	Cooperation	Overcooked	-	✗	Txt
HLA [105]	Cooperation	Overcooked	GPT-3.5&LLaMA2	✗	Txt
S-Agents [99]	Cooperation	Minecraft	GPT-4	✗	Txt
HAC [67]	Cooperation	Minecraft	GPT-4V	✗	Txt & Img
CoELA [35]	Cooperation	TDW-T&WAH	GPT-4	✗	Txt & Img
GenerativeAgents [59]	Human Simulation	SmallVille	GPT-3.5	✗	Txt
HumanoidAgents [134]	Human Simulation	Social	GPT-3.5	✗	Txt
LyfeAgent [121]	Human Simulation	Lyfe Game	GPT-3.5	✗	Txt
AgentSims [135]	Human Simulation	AgentSims	-	✗	Txt
CivRealm [56]	Civil. Simulation	Civilization	-	✗	Txt
ZeroShotPlanner [98]	Embodied Simulation	VirtualHome	GPT-3	✗	Txt
LLMPlanner [36]	Embodied Simulation	ALFRED	GPT-3	✗	Txt & Img
E2WM [110]	Embodied Simulation	VirtualHome	LLaMA-13B	✓	Txt
Octopus [45]	Embodied Simulation	Behavior-1K	CLIP & MPT-7B	✓	Txt & Img
Voyager [65]	Craft & Explore	Minecraft	GPT-4	✗	Txt
DEPS [38]	Craft & Explore	Minecraft	MineCLIP & GPT-4	✗	Txt & Img
GTIM [66]	Craft & Explore	Minecraft	GPT-3.5	✗	Txt
JARVIS-1 [39]	Craft & Explore	Minecraft	MineCLIP & GPT4	✗	Txt & Img
Plan4MC [101]	Craft & Explore	Minecraft	GPT-3.5	✗	Txt & Img
RL-GPT [106]	Craft & Explore	Minecraft	GPT-4	✗	Txt & Img
MineDoJo [41]	Craft & Explore	Minecraft	MineCLIP	✓	Txt & Img
LLaMARider [111]	Craft & Explore	Minecraft	LLaMA-2-70B	✓	Txt & Img
SteveEye [136]	Craft & Explore	Minecraft	CLIP & LLaMA2	✓	Txt & Img
CreativeAgent [44]	Craft & Explore	Minecraft	GPT-4V	✗	Txt & Img
MCReward [44]	Craft & Explore	Minecraft	GPT-4	✗	Txt & Img
ELLM [40]	Craft & Explore	Crafter	Codex	✗	Txt & Img
SPRING [94]	Craft & Explore	Crafter	GPT-4	✗	Txt & Img
AdaRefiner [95]	Craft & Explore	Crafter	LLaMA2 & GPT-4	✓	Txt & Img
OMNI [93]	Craft & Explore	Crafter	GPT-3	✗	Txt & Img
PlayDoom [43]	Others	Doom	GPT-4V	✗	Txt & Img
GATO [46]	Others	Atari	GATO	✓	Img
Motif [117]	Others	NetHack	LlaMA-2	✗	Txt

3.2 Communication Games

Communication (or conversational) games revolve through the turns of communication, negotiation, deduction and deception among multiple players. The challenge of communication games lies in inferring others' intention behind ambiguous or misleading language utterances, and hiding one's own intention if necessary.

Werewolf: The game pits two groups against each other, *i.e.*, werewolves and non-werewolves (villagers, witch, guard and seer), and alternates between night phases, where werewolves secretly attack, and day phases, where survivors discuss and vote to eliminate suspects. The witch, guard, and seer each possess unique abilities. Xu et al. [28] propose to retrieve and reflect on historical communications for enhancement, and observe that GPT-3.5 demonstrates strategic behaviors such as trust, confrontation, camouflage, and leadership. Xu et al. [125] employ a RL policy to select the optimal action from among the diverse actions generated by LLMs, aiming to overcome the LLMs' prior preference for specific actions. Wu et al. [115] introduce a RL policy to generate the next action by taking as input the reasoning generated by the LLM, and employ another LLM to generate descriptions aligned to the action.

Avalon: The game progresses through rounds of discussion and voting to decide who participates in the quests. The goal for the good team is to successfully complete quests, while the bad team aims to secretly sabotage these quests or identify the role of Merlin, who knows the identities of the bad players. Light et al. [126] suggest that GPT-3.5 struggles to formulate and execute simple strategies and sometimes reveals its own bad identity. Wang et al. [52] introduce a reasoning approach that takes into account first-order and second-order perspective shifts to combat pervasive misinformation. To combat hallucination, Shi et al. [127] propose to generate reasoning substeps in a code format that are interpreted as actions subsequently.

Diplomatic games: Diplomacy is the first diplomatic board game from the 1950s where players assume the roles of seven powers striving to conquer Europe during WW1. Each turn is marked by private negotiations, trust-building, and tactical coordination among players. Cicero [51] is a human-level agent in Diplomacy that integrates a RL policy for planning and a BART [147] model conditioned on the plan for generating consistent negotiation messages; WarAgent [109] simulates the participating countries, decisions, and consequences in WW I and WW II; CosmoAgent [128] mimics the communication, conflicts, and cooperation among various universal civilizations.

Others: Studies have demonstrated the game-playing abilities of LLMs in various games, including SpyGame (Who is Spy) [130; 129], Ask-Guess [130; 129], Tofu Kingdom [130], and Murder Mystery Games [148], known as Jubensha in Chinese.

3.3 Competition Games

Competition games, governed by strict rules, challenge agents with varied-level opponents, demanding advanced reasoning and skills. Competition games serves as benchmarks for evaluating reasoning and planning abilities of LLMGAs directly against human players. Reaching human-level performance is a crucial achievement that highlights the agent's prowess in complex decision-making and strategic implementation.

StarCraft II: StarCraft II is a real-time strategy game in which players are tasked with gathering resources, building bases, creating armies, and engaging in combats to defeat the opponent. Ma et al. [29] introduce TextStarCraft II, a natural language interface that enables LLMs to play StarCraft II and Chain-of-Summarization for efficient reasoning and decision-making; SwarmBrain [131] introduce an Overmind Intelligence Matrix for high-level strategic planning and a Swarm ReflexNet for rapid tactical responses. These LLM-based agents exhibit comparable performance against the game's built-in AI at high difficulty levels.

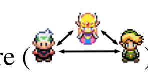
Pokémon Battle: Pokémon battles are turn-based tactical games, with two players each sending out one Pokémon and choosing either to attack or switch Pokémon each turn. Hu et al. [30] introduce an environment that enables LLMs to play Pokémon battles and a human-level agent PokéLLMon that consumes instant feedback to iteratively refine the policy, retrieves external knowledge to combat hallucination, and generates consistent actions to alleviate the panic switching problem caused by CoT [81] reasoning.

Chess: Feng et al. [55] introduce a large-scale chess gameplay dataset stored in Portable Game Notation format [149] and ChessGPT fine-tuned on mixed chess and language datasets to support board state evaluation and chess playing; Toshniwal et al. [150] and Li et al. [132] discover that LMs trained to predict next move in chess are capable of tracking the state of the board given a move sequence, *i.e.*, LMs are capable of playing blindfolded. This suggests that LMs do not merely memorize surface statistics but also learn a causal model of the sequence-generating process.

Poker: In Texas Hold'em, Gupta et al. [151] observe that GPT-4 plays like an advanced yet aggressive player who raises with a wide range of hands pre-flop, avoids limping, and exhibits unconventional play; PokerGPT [53] demonstrates that OPT-1.3B [152] with supervised fine-tuning and RLHF [2] can achieve comparable performance to a RL-based method Alphaholdem [153] with significantly less training cost: 9.5 GPU hours compared to Alphaholdem's 580 GPU hours. Guo et al. [54] and Zhang et al. [61] demonstrate that prompting LLMs to predict opponents' thoughts, known as Theory of Mind [90; 91], results in significant improvements in Texas Hold'em, BlackJack and Leduc Hold'em [11].

3.4 Cooperation Games

Cooperation among individuals can enhance the efficiency and effectiveness of task accomplishment. There are primarily three types of cooperative tasks in games: (1) **Cooperative cooking** [102; 100; 154] requires agents collaborate to cook and deliver as many dishes as possible within the given time. To prepare an onion soup in Overcooked-AI [102], two agents need to load three onions into a cooker, starting a cooking process that lasts 20 time steps, and transfer the soup to a plate for delivery; (2) **Embodied household cooperation** [155; 156] requires agents to collaboratively accomplish tasks like transporting as many objects as possible to the goal position in embodied environments with partial observation [107; 157]; (3) **Cooperative crafting** [99; 100] **& exploration** [67] in Minecraft can be accelerated through cooperation between multiple agents. Existing cooperative game settings can be categorized into decentralized and centralized cooperation.

Decentralized cooperation: A decentralized structure is a democratic structure () where there is no central task dispatcher. In Overcooked, the ability to infer the partner's intent and next action based on the its historical actions, known as Theory-of-Mind, is crucial to prevent conflicts. Agashe et al. [92] show that GPT-4 is able to recognize and offer assistance to partners in need, and show robustness in adjusting to different partners. ProAgent [87] introduces a belief correction module to rectify incorrect beliefs on partners and consistently outperforms RL approaches [114; 31; 158]. Moreover, HLA [105] integrates a proficient LLM and a lightweight LLM to balance efficacy and efficiency in real-time human-agent interaction; In partially-observable embodied environments, CoELA [35] introduce an efficient communication module to determine what and when to communicate, exhibiting better performance compare to MCTS-based and rule-based planners on Watch-and-Help [155] and TDW Transport tasks [156].

Centralized cooperation: In Minecraft, S-agents [99] and MindAgents [100] adopts a centralized dispatcher/planner to decompose a challenging goal into subtasks and dispatches them to agents for execution, forming a hierarchical architecture. HAS [67] introduces an auto-organizing mechanism to dynamically adjust key roles and action groups during cooperation, and an intra-communication mechanism to ensure efficient collaboration.

3.5 Simulation Games

Simulation games provide simulated environments for real-world events or scenarios, enabling players to experience realistic interactions and decision-making in open-ended game playing. Existing studies can be categorized as human & social simulation, civilization simulation and embodied simulation.

Human and social simulation: Generative Agents [59] marks the first LLM-based human simulation experiment that leverages LLMs' prior knowledge to simulate human-like daily life and social activities. Specifically, GPT-3.5 assumes the roles of 25 generative agents with unique persona and social relationship, residing in a virtual small town. A cognitive architecture is introduced to support agents in remembering, retrieving, reflecting, planning, and acting within dynamic environments. During the two-day simulation, emergent behaviors like exchanging information, forming new relationships and coordinating joint activities are observed.

On the basis of Generative Agents, Humanoid Agents [134] further considers the effects of states like basic needs (e.g., hunger, health, and energy), emotions, and closeness in relationships on agents' behavior generation; For other simulation environments, AgentSims [135] is a programmable and extendable environment; LyfeGame [121] is an 3D virtual small town in Japan. Three experimental scenarios are designed to assess the social behaviors of LLM-based agents, including a murder mystery, a high school activity fair, and a patient-in-distress scenario.

Civilization simulation: CivRealm [56] is a game environment based on Civilization [32], where each player governs a civilization simulating the progress of human history. As an open-ended game, it features diverse victory conditions, requiring players to strategically develop the economy, military, diplomacy, culture, and technology of their civilizations. Mastaba [56] introduces an advisor and an AutoGPT [159]-style worker, where the advisor aids in generating context-specific objectives while the workers handle the execution of these goals through generated actions. Experiments show that the advisor brings an advantage at the early game stage, yet the advantage diminishes as the game progresses.

Embodied simulation: In simulated 3D environments, embodied agents perceives their surroundings from egocentric perception similar to human and engage with realistic objects to carry out a wide range of tasks by following instructions like "Rinse off a mug and place it in the coffee maker". Existing benchmarks include AI2-THOR [160], Virtual Home [107], ALFRED [140], iGibson [161], Habitat [162], ThreeDWorld [156], Behavior [163] and Behavior-1K [164]. Existing approaches [36; 98; 108; 110] primarily adopt LLMs as planners to decompose an instruction into action plans. Specifically, ZeroShotPlanner [98] prompts LLMs in zero-shot manner for planning; SayCan [108] uses a learned affordance function to assist LLMs in selecting valid actions during planning; LLMPlanner [36] adopts an KNN retriever to select few-shot examples and dynamically re-plan based on the observation in the current environment; E2WM [110] fine-tunes an LLM with embodied experience collected through action space search and random exploration to enhance the understanding of the environments.

3.6 Crafting & Exploration Games

Minecraft and Crafter are two game environments that have been widely studied for game agents with a focus on crafting & exploration. Minecraft [14] is a 3D sandbox game that offer players the great freedom to traverse a world made up of blocky, pixelated landscapes, facilitated by the procedurally generated worlds. The resouce-based crafting system enables players to transform collected materials into tools, build elaborate structures and complex machines. Crafter [122] is a 2D open-world game that mirrors the survival mode of Minecraft. It challenges players to manage their resources carefully to ensure sufficient water, food, and rest, while also defending against threats like zombies. The game's world is also procedurally generated for the exploration purpose, and it includes 22 tasks for players to accomplish.

Existing agents can be divided as goal-conditioned agents that implement the task given an instruction (crafting), or autonomous exploration agents that navigate within the open-world based on self-determined objectives (exploration).

Crafting: The key challenge in crafting tasks lies in their complexity: agents must gather diverse materials scattered across the world and understand intricate recipes and the sequential steps involved. Consequently, planning is widely employed to address crafting tasks. Existing agent design such as DEPS [38], GITM [66], JARVIS-1 [39], Plan4MC [101], RL-GPT [106] and S-agents [99] mainly follow a paradigm that adopts LLMs as a planner to decompose the goal into subgoals and further generate action plans for each sub-goals. Specifically, DEPS introduce error correction on initial plans by integrating description of the plan execution and self-explanation of feedback when encountering failures; GITM [66] leverages external knowledges like item crafting/smelting recipes, and is equipped with a long-term memory to maintain common reference plans for encountered objectives; JARVIS-1 [39] chains MineCLIP [41] and an LLM together to perceive multimodal input and utilizes a multimodal memory to store experiences; Plan4MC [101] and RL-GPT [106] integrate the LLM planner with a low-level RL policy for action execution; S-agents [99] and HAS [67] dispatches subtasks to multiple agents for cooperatively task execution;

Exploration: Navigating through procedurally generated world without specific goals can overwhelm agents with numerous possible actions. Previous works leverage curriculum learning [165] to identify suitable tasks while now LLMs can directly act as goal generators. In Minecraft, Voyager [65] adopts an automatic curriculum in a self-directed way [166], *i.e.*, it asks LLM to generate goals that adapts to the agent's current state, inventory, acquired skills, and environment. In Crafter, OMNI [93] utilizes LLMs to determine interesting tasks for curriculum design, overcoming the previous challenge of quantifying "interest". ELLM [40], SPRING [94] and AdaRefiner [95] prompt LLMs to generate goals for agents. Specifically, ELLM [40] queries LLMs for next goals given an agent's current context, and rewards agents for accomplishing those suggestions in the sparse-reward setting; SPRING [94] uses LLMs to summarize useful knowledge from the Crafter paper [122] and progressively prompts the LLM to generate next action; On the basis of ELLM, AdaRefiner [95] cascades a learnable lightweight LLM with fixed LLMs for better goal plan generation.

3.7 Evaluation

The evaluation metrics for game agents vary across different games. In Table 2, we summarize the metrics for several representative games. For games with specific task instructions, such as ALFWorld, ScienceWorld, BabyAI, RDR2, ALFRED, VirtualHome, and crafting tasks in Minecraft and Crafter, the task success rate is usually adopted as the primary metric; for competition games, win rate, game score, and Elo rating are common metrics; for communication games that separate players into adversarial teams such as Werewolf and Avalone, win rate can be used as the metric. For human/social simulation experiments, human evaluators are typically recruited to assess the believability of behaviors exhibited by LLM-based agents.

Table 2: Evaluation metric used in representative games

Game	Metric
Jericho	Game score [123]
ALFWorld	Task success rate [80; 96]
ScienceWorld	Task score [97]
BabyAI/BabyAI-Text	Task success rate [93; 113]
RDR2	Task success rate [34]
Werewolf	Win rate [115; 28; 125], Voting accuracy [115]
Avalone	Win rate [52; 127; 126]
Diplomacy	Player score [51]
StarCraft II	Win rate [29; 131]
Pokémon Battles	Win rate [30], Battle score [30]
Chess	Elo rating [55], Move score [55]
Poker	Win rate [54; 53], # of win/loss chips [54], mbb/hand [53]
Overcooked	Reward value [92; 87], Success rate [100; 105]
Human Simulation	Human evaluation [59; 134]
Civilization	Task success rate [56], Game score [56], # of techs & units [56]
ALFRED	Task success rate [36]
VirtualHome	Task success rate [98; 167], Executability [98; 167]
Minecraft	Task success rate [38; 65; 66], Map coverage [65], # of items [65]
Crafter	Task success rate [40; 93]

4 Conclusion and Future Directions

In this paper, we conduct a systematic literature review of existing studies on LLMGAs, examining two primary aspects: (1) Construction of LLMGAs, where we elaborate on six essential components, including perception, memory, thinking, role-playing, action, and learning; (2) LLMGAs in six game categories, including adventure, communication, competition, cooperation, simulation, and crafting & exploration, where we detail the game environments and common strategies adopted by game agents associated with each type. Finally, we identify three potential future directions for this new research field:

Grounding LLMs in environments: LLMs pre-trained only on text corpus data are not grounded in real environments, *i.e.*, they are not really aware of the consequences of their generations on physical process [168]. Consequently, ungrounded LLMs generate inadmissible actions [108], exhibit a gap between high-level intentions and intricate game control [34], especially in the absence of visual perception abilities [29] and feedback, and thus largely rely on manually-designed prompts. Existing efforts have been made toward grounding LLMs through multimodal perception [45], the adopt of external affordance functions [108], feedback from environments [30], and experiencing in environments [110; 40]. However, the current progress in grounding techniques remains limited and fall shorts for the requirements of real-world application [169]. Games, serving as controllable and safe environments, are ideal testbeds for developing grounding techniques that can make LLMs more inline with sophisticated environments.

Knowledge discovery through game-playing: Current studies [65; 30; 29] primarily remain at the stage of utilizing the pre-existing knowledge encoded in LLMs for game-playing. Although some studies propose to leverage game-playing experiences [110; 111; 40] to ground and enhance LLM-based agents, these agents are still incapable of extracting underlying knowledge below the surfaces of observations and experience. Knowledge discovery is not simply about learning to act effectively, but to understand fundamental principles and causal model of gameplay mechanisms just like human. We believe that gameplay mechanisms with complex extrinsic knowledge are essential testbeds for designing such agents, and knowledge discovery via experiencing in environments might represent a critical step toward the pursuit of AGI.

Agent society simulation: The social simulation experiment of Generative Agents [59] has demonstrated that LLM-based agents are promising for believable simulacra of human. Emergent human-like social behaviors are observed like information diffusion, forming new relationship and coordination for social activities, with the support of a novel cognitive architecture. However, as human beings are far more sophisticated, with complicated mental processes, emotional depth, and advanced social skills, it would be an intriguing avenue for future research to develop better cognitive architectures and more nuanced simulations of social interactions and cooperation [170] in realistic game environments to foster deeper understanding and representation of complex human interactions.

Acknowledgements

Authors would like to thank Yunxiang Yan and Vijayraj Shanmugaraj for their assistance in collecting papers. This research is partially sponsored by the NSF CISE grants 2302720, 2312758, 2038029, a GTRI PhD Fellowship, an IBM faculty award, and a grant from CISCO Edge AI program.

References

[1] Smith, L., M. Gasser. The development of embodied cognition: Six lessons from babies. *Artificial life*, 11(1-2):13–29, 2005.

[2] Ouyang, L., J. Wu, X. Jiang, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744, 2022.

[3] Achiam, J., S. Adler, S. Agarwal, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.

[4] Team, G., R. Anil, S. Borgeaud, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.

[5] Sumers, T. R., S. Yao, K. Narasimhan, et al. Cognitive architectures for language agents. *arXiv preprint arXiv:2309.02427*, 2023.

[6] Delétang, G., A. Ruoss, P.-A. Duquenne, et al. Language modeling is compression. *arXiv preprint arXiv:2309.10668*, 2023.

[7] Goertzel, B., C. Pennachin. *Artificial general intelligence*, vol. 2. Springer, 2007.

[8] Goertzel, B. Artificial general intelligence: concept, state of the art, and future prospects. *Journal of Artificial General Intelligence*, 5(1):1, 2014.

[9] IBM. Deep blue. https://www.ibm.com/history/deep-blue.

[10] Silver, D., A. Huang, C. J. Maddison, et al. Mastering the game of go with deep neural networks and tree search. *nature*, 529(7587):484–489, 2016.

[11] Southey, F., M. P. Bowling, B. Larson, et al. Bayes' bluff: Opponent modelling in poker. *arXiv preprint arXiv:1207.1411*, 2012.

[12] Bellemare, M. G., Y. Naddaf, J. Veness, et al. The arcade learning environment: An evaluation platform for general agents. *Journal of Artificial Intelligence Research*, 47:253–279, 2013.

[13] Vinyals, O., T. Ewalds, S. Bartunov, et al. Starcraft ii: A new challenge for reinforcement learning. *arXiv preprint arXiv:1708.04782*, 2017.

[14] Mojang Studios. Minecraft. https://www.minecraft.net/en-us.

[15] Dota 2. https://www.dota2.com/home.

[16] Silver, D., J. Schrittwieser, K. Simonyan, et al. Mastering the game of go without human knowledge. *nature*, 550(7676):354–359, 2017.

[17] OpenAI. Openai five. https://openai.com/research/openai-five, 2018. Accessed on: yyyy-mm-dd.

[18] DeepMind. Alphastar: Mastering the real-time strategy game starcraft ii. https://deepmind.google/discover/blog/alphastar-mastering-the-real-time-strategy-game-starcraft-ii, 2019.

[19] Zhao, W. X., K. Zhou, J. Li, et al. A survey of large language models. *arXiv preprint arXiv:2303.18223*, 2023.

[20] Chang, Y., X. Wang, J. Wang, et al. A survey on evaluation of large language models. *ACM Transactions on Intelligent Systems and Technology*, 2023.

[21] Kaddour, J., J. Harris, M. Mozes, et al. Challenges and applications of large language models. *arXiv preprint arXiv:2307.10169*, 2023.

[22] Xi, Z., W. Chen, X. Guo, et al. The rise and potential of large language model based agents: A survey. *arXiv preprint arXiv:2309.07864*, 2023.

[23] Wang, L., C. Ma, X. Feng, et al. A survey on large language model based autonomous agents. *arXiv preprint arXiv:2308.11432*, 2023.

[24] Gao, C., X. Lan, N. Li, et al. Large language models empowered agent-based modeling and simulation: A survey and perspectives. *arXiv preprint arXiv:2312.11970*, 2023.

[25] Gallotta, R., G. Todd, M. Zammit, et al. Large language models and games: A survey and roadmap. *arXiv preprint arXiv:2402.18659*, 2024.

[26] Sweetser, P. Large language models and video games: A preliminary scoping review. *arXiv preprint arXiv:2403.02613*, 2024.

[27] Infocom. Zork I. http://ifdb.tads.org/viewgame?id=0dbnusxunq7fw5ro, 1980.

[28] Xu, Y., S. Wang, P. Li, et al. Exploring large language models for communication games: An empirical study on werewolf. *arXiv preprint arXiv:2309.04658*, 2023.

[29] Ma, W., Q. Mi, X. Yan, et al. Large language models play starcraft ii: Benchmarks and a chain of summarization approach. *arXiv preprint arXiv:2312.11865*, 2023.

[30] Hu, S., T. Huang, L. Liu. Pokéllmon: A human-parity agent for pokémon battles with large language models, 2024.

[31] Li, Y., S. Zhang, J. Sun, et al. Cooperative open-ended learning framework for zero-shot coordination. *arXiv preprint arXiv:2302.04831*, 2023.

[32] Freeciv-web contributors. Freeciv-web. https://github.com/freeciv/freeciv-web, 2023.

[33] PrismarineJS. Mineflayer: Create minecraft bots with a powerful, stable, and high level javascript api. https://github.com/PrismarineJS/mineflayer, 2013.

[34] Tan, W., Z. Ding, W. Zhang, et al. Towards general computer control: A multimodal agent for red dead redemption ii as a case study. *arXiv preprint arXiv:2403.03186*, 2024.

[35] Zhang, H., W. Du, J. Shan, et al. Building cooperative embodied agents modularly with large language models. *arXiv preprint arXiv:2307.02485*, 2023.

[36] Song, C. H., J. Wu, C. Washington, et al. Llm-planner: Few-shot grounded planning for embodied agents with large language models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 2998–3009. 2023.

[37] Radford, A., J. W. Kim, C. Hallacy, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PMLR, 2021.

[38] Wang, Z., S. Cai, A. Liu, et al. Describe, explain, plan and select: Interactive planning with large language models enables open-world multi-task agents. *arXiv preprint arXiv:2302.01560*, 2023.

[39] —. Jarvis-1: Open-world multi-task agents with memory-augmented multimodal language models. *arXiv preprint arXiv:2311.05997*, 2023.

[40] Du, Y., O. Watkins, Z. Wang, et al. Guiding pretraining in reinforcement learning with large language models. In *International Conference on Machine Learning*, pages 8657–8677. PMLR, 2023.

[41] Fan, L., G. Wang, Y. Jiang, et al. Minedojo: Building open-ended embodied agents with internet-scale knowledge. *Advances in Neural Information Processing Systems*, 35:18343–18362, 2022.

[42] Mokady, R., A. Hertz, A. H. Bermano. Clipcap: Clip prefix for image captioning. *arXiv preprint arXiv:2111.09734*, 2021.

[43] de Wynter, A. Will gpt-4 run doom? *arXiv preprint arXiv:2403.05468*, 2024.

[44] Zhang, C., P. Cai, Y. Fu, et al. Creative agents: Empowering agents with imagination for creative tasks. *arXiv preprint arXiv:2312.02519*, 2023.

[45] Yang, J., Y. Dong, S. Liu, et al. Octopus: Embodied vision-language programmer from environmental feedback. *arXiv preprint arXiv:2310.08588*, 2023.

[46] Reed, S., K. Zolna, E. Parisotto, et al. A generalist agent. *arXiv preprint arXiv:2205.06175*, 2022.

[47] Zheng, S., Y. Feng, Z. Lu, et al. Steve-eye: Equipping llm-based embodied agents with visual perception in open worlds. In *The Twelfth International Conference on Learning Representations*. 2023.

[48] Infocom. Zork III. http://ifdb.tads.org/viewgame?id=vrsot1zgy1wfcdru, 1982.

[49] Wang, R., P. Jansen, M.-A. Côté, et al. Scienceworld: Is your agent smarter than a 5th grader? *arXiv preprint arXiv:2203.07540*, 2022.

[50] Shridhar, M., X. Yuan, M.-A. Côté, et al. Alfworld: Aligning text and embodied environments for interactive learning. *arXiv preprint arXiv:2010.03768*, 2020.

[51] (FAIR)†, M. F. A. R. D. T., A. Bakhtin, N. Brown, et al. Human-level play in the game of diplomacy by combining language models with strategic reasoning. *Science*, 378(6624):1067–1074, 2022.

[52] Wang, S., C. Liu, Z. Zheng, et al. Avalon's game of thoughts: Battle against deception through recursive contemplation. *arXiv preprint arXiv:2310.01320*, 2023.

[53] Huang, C., Y. Cao, Y. Wen, et al. Pokergpt: An end-to-end lightweight solver for multi-player texas hold'em via large language model. *arXiv preprint arXiv:2401.06781*, 2024.

[54] Guo, J., B. Yang, P. Yoo, et al. Suspicion-agent: Playing imperfect information games with theory of mind aware gpt-4. *arXiv preprint arXiv:2309.17277*, 2023.

[55] Feng, X., Y. Luo, Z. Wang, et al. Chessgpt: Bridging policy learning and language modeling. *Advances in Neural Information Processing Systems*, 36, 2024.

[56] Qi, S., S. Chen, Y. Li, et al. Civrealm: A learning and reasoning odyssey in civilization for decision-making agents. *arXiv preprint arXiv:2401.10568*, 2024.

[57] Atkinson, R. C., R. M. Shiffrin. Human memory: A proposed system and its control processes. In *Psychology of learning and motivation*, vol. 2, pages 89–195. Elsevier, 1968.

[58] Nuxoll, A. M., J. E. Laird. Extending cognitive architecture with episodic memory. In *AAAI*, pages 1560–1564. 2007.

[59] Park, J. S., J. O'Brien, C. J. Cai, et al. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, pages 1–22. 2023.

[60] Tuyls, J., S. Yao, S. Kakade, et al. Multi-stage episodic control for strategic exploration in text games. *arXiv preprint arXiv:2201.01251*, 2022.

[61] Zhang, W., K. Tang, H. Wu, et al. Agent-pro: Learning to evolve via policy-level reflection and optimization. *arXiv preprint arXiv:2402.17574*, 2024.

[62] Lindes, P., J. E. Laird. Toward integrating cognitive linguistics and cognitive language processing. In *Proceedings of the 14th International Conference on Cognitive Modeling (ICCM)*. 2016.

[63] Shinn, N., B. Labash, A. Gopinath. Reflexion: an autonomous agent with dynamic memory and self-reflection. *arXiv preprint arXiv:2303.11366*, 2023.

[64] Lum, J. A., G. Conti-Ramsden, D. Page, et al. Working, declarative and procedural memory in specific language impairment. *cortex*, 48(9):1138–1154, 2012.

[65] Wang, G., Y. Xie, Y. Jiang, et al. Voyager: An open-ended embodied agent with large language models. *arXiv preprint arXiv:2305.16291*, 2023.

[66] Zhu, X., Y. Chen, H. Tian, et al. Ghost in the minecraft: Generally capable agents for open-world enviroments via large language models with text-based knowledge and memory. *arXiv preprint arXiv:2305.17144*, 2023.

[67] Zhao, Z., K. Chen, D. Guo, et al. Hierarchical auto-organizing system for open-ended multi-agent navigation. *arXiv preprint arXiv:2403.08282*, 2024.

[68] Uludağlı, M. Ç., K. Oğuz. Non-player character decision-making in computer games. *Artificial Intelligence Review*, 56(12):14159–14191, 2023.

[69] Zhu, A., L. Martin, A. Head, et al. Calypso: Llms as dungeon master's assistants. In *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, vol. 19, pages 380–390. 2023.

[70] Wang, Z. M., Z. Peng, H. Que, et al. Rolellm: Benchmarking, eliciting, and enhancing role-playing abilities of large language models. *arXiv preprint arXiv:2310.00746*, 2023.

[71] Jiang, H., X. Zhang, X. Cao, et al. Personallm: Investigating the ability of gpt-3.5 to express personality traits and gender differences. *arXiv preprint arXiv:2305.02547*, 2023.

[72] Li, C., Z. Leng, C. Yan, et al. Chatharuhi: Reviving anime character in reality via large language model. *arXiv preprint arXiv:2308.09597*, 2023.

[73] Salemi, A., S. Mysore, M. Bendersky, et al. Lamp: When large language models meet personalization. *arXiv preprint arXiv:2304.11406*, 2023.

[74] Li, G., H. Hammoud, H. Itani, et al. Camel: Communicative agents for" mind" exploration of large language model society. *Advances in Neural Information Processing Systems*, 36, 2024.

[75] Croissant, M., M. Frister, G. Schofield, et al. An appraisal-based chain-of-emotion architecture for affective language model game agents. *arXiv preprint arXiv:2309.05076*, 2023.

[76] Huang, J.-t., M. H. Lam, E. J. Li, et al. Emotionally numb or empathetic? evaluating how llms feel using emotionbench. *arXiv preprint arXiv:2308.03656*, 2023.

[77] Tu, Q., S. Fan, Z. Tian, et al. Charactereval: A chinese benchmark for role-playing conversational agent evaluation. *arXiv preprint arXiv:2401.01275*, 2024.

[78] Shao, Y., L. Li, J. Dai, et al. Character-llm: A trainable agent for role-playing. *arXiv preprint arXiv:2310.10158*, 2023.

[79] Huang, J., K. C.-C. Chang. Towards reasoning in large language models: A survey. *arXiv preprint arXiv:2212.10403*, 2022.

[80] Yao, S., J. Zhao, D. Yu, et al. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2022.

[81] Wei, J., X. Wang, D. Schuurmans, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35:24824–24837, 2022.

[82] Kojima, T., S. S. Gu, M. Reid, et al. Large language models are zero-shot reasoners. *Advances in neural information processing systems*, 35:22199–22213, 2022.

[83] Yao, S., D. Yu, J. Zhao, et al. Tree of thoughts: Deliberate problem solving with large language models. *arXiv preprint arXiv:2305.10601*, 2023.

[84] Wang, X., J. Wei, D. Schuurmans, et al. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*, 2022.

[85] Besta, M., N. Blach, A. Kubicek, et al. Graph of thoughts: Solving elaborate problems with large language models. *arXiv preprint arXiv:2308.09687*, 2023.

[86] Madaan, A., N. Tandon, P. Gupta, et al. Self-refine: Iterative refinement with self-feedback. *Advances in Neural Information Processing Systems*, 36, 2024.

[87] Zhang, C., K. Yang, S. Hu, et al. Proagent: Building proactive cooperative ai with large language models. *arXiv preprint arXiv:2308.11339*, 2023.

[88] Kim, G., P. Baldi, S. McAleer. Language models can solve computer tasks. *Advances in Neural Information Processing Systems*, 36, 2024.

[89] Hu, S., T. Huang, F. İlhan, et al. Large language model-powered smart contract vulnerability detection: New perspectives. *arXiv preprint arXiv:2310.01152*, 2023.

[90] Frith, C., U. Frith. Theory of mind. *Current biology*, 15(17):R644–R645, 2005.

[91] Kosinski, M. Theory of mind may have spontaneously emerged in large language models. *arXiv preprint arXiv:2302.02083*, 2023.

[92] Agashe, S., Y. Fan, X. E. Wang. Evaluating multi-agent coordination abilities in large language models. *arXiv preprint arXiv:2310.03903*, 2023.

[93] Zhang, J., J. Lehman, K. Stanley, et al. Omni: Open-endedness via models of human notions of interestingness. *arXiv preprint arXiv:2306.01711*, 2023.

[94] Wu, Y., S. Y. Min, S. Prabhumoye, et al. Spring: Studying papers and reasoning to play games. *Advances in Neural Information Processing Systems*, 36, 2024.

[95] Zhang, W., Z. Lu. Adarefiner: Refining decisions of language models with adaptive feedback, 2023.

[96] Prasad, A., A. Koller, M. Hartmann, et al. Adapt: As-needed decomposition and planning with language models. *arXiv preprint arXiv:2311.05772*, 2023.

[97] Lin, B. Y., Y. Fu, K. Yang, et al. Swiftsage: A generative agent with fast and slow thinking for complex interactive tasks. *Advances in Neural Information Processing Systems*, 36, 2024.

[98] Huang, W., P. Abbeel, D. Pathak, et al. Language models as zero-shot planners: Extracting actionable knowledge for embodied agents. In *International Conference on Machine Learning*, pages 9118–9147. PMLR, 2022.

[99] Chen, J., Y. Jiang, J. Lu, et al. S-agents: self-organizing agents in open-ended environment. *arXiv preprint arXiv:2402.04578*, 2024.

[100] Gong, R., Q. Huang, X. Ma, et al. Mindagent: Emergent gaming interaction. *arXiv preprint arXiv:2309.09971*, 2023.

[101] Yuan, H., C. Zhang, H. Wang, et al. Plan4mc: Skill reinforcement learning and planning for open-world minecraft tasks. *arXiv preprint arXiv:2303.16563*, 2023.

[102] Carroll, M., R. Shah, M. K. Ho, et al. On the utility of learning about humans for human-ai coordination. *Advances in neural information processing systems*, 32, 2019.

[103] Microsoft Research. First textworld problems: The competition using text-based games to advance capabilities of ai agents, 2019.

[104] Hausknecht, M., P. Ammanabrolu, M.-A. Côté, et al. Interactive fiction games: A colossal adventure. In *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, pages 7903–7910. 2020.

[105] Liu, J., C. Yu, J. Gao, et al. Llm-powered hierarchical language agent for real-time human-ai coordination. *arXiv preprint arXiv:2312.15224*, 2023.

[106] Liu, S., H. Yuan, M. Hu, et al. Rl-gpt: Integrating reinforcement learning and code-as-policy. *arXiv preprint arXiv:2402.19299*, 2024.

[107] Puig, X., K. Ra, M. Boben, et al. Virtualhome: Simulating household activities via programs. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 8494–8502. 2018.

[108] Ahn, M., A. Brohan, N. Brown, et al. Do as i can, not as i say: Grounding language in robotic affordances. *arXiv preprint arXiv:2204.01691*, 2022.

[109] Hua, W., L. Fan, L. Li, et al. War and peace (waragent): Large language model-based multi-agent simulation of world wars. *arXiv preprint arXiv:2311.17227*, 2023.

[110] Xiang, J., T. Tao, Y. Gu, et al. Language models meet world models: Embodied experiences enhance language models. *Advances in neural information processing systems*, 36, 2024.

[111] Feng, Y., Y. Wang, J. Liu, et al. Llama rider: Spurring large language models to explore the open world. *arXiv preprint arXiv:2310.08922*, 2023.

[112] Li, S., X. Puig, C. Paxton, et al. Pre-trained language models for interactive decision-making. *Advances in Neural Information Processing Systems*, 35:31199–31212, 2022.

[113] Carta, T., C. Romac, T. Wolf, et al. Grounding large language models in interactive environments with online reinforcement learning. *arXiv preprint arXiv:2302.02662*, 2023.

[114] Schulman, J., F. Wolski, P. Dhariwal, et al. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

[115] Wu, S., L. Zhu, T. Yang, et al. Enhance reasoning for large language models in the game werewolf. *arXiv preprint arXiv:2402.02330*, 2024.

[116] Kwon, M., S. M. Xie, K. Bullard, et al. Reward design with language models. *arXiv preprint arXiv:2303.00001*, 2023.

[117] Klissarov, M., P. D'Oro, S. Sodhani, et al. Motif: Intrinsic motivation from artificial intelligence feedback. *arXiv preprint arXiv:2310.00166*, 2023.

[118] Li, H., X. Yang, Z. Wang, et al. Auto mc-reward: Automated dense reward design with large language models for minecraft. *arXiv preprint arXiv:2312.09238*, 2023.

[119] Xie, T., S. Zhao, C. H. Wu, et al. Text2reward: Automated dense reward function generation for reinforcement learning. *arXiv preprint arXiv:2309.11489*, 2023.

[120] Ma, Y. J., W. Liang, G. Wang, et al. Eureka: Human-level reward design via coding large language models. *arXiv preprint arXiv:2310.12931*, 2023.

[121] Kaiya, Z., M. Naim, J. Kondic, et al. Lyfe agents: Generative agents for low-cost real-time social interactions. *arXiv preprint arXiv:2310.02172*, 2023.

[122] Hafner, D. Benchmarking the spectrum of agent capabilities. *arXiv preprint arXiv:2109.06780*, 2021.

[123] Yao, S., R. Rao, M. Hausknecht, et al. Keep calm and explore: Language models for action generation in text-based games. *arXiv preprint arXiv:2010.02903*, 2020.

[124] Tsai, C. F., X. Zhou, S. S. Liu, et al. Can large language models play text games well? current state-of-the-art and open questions. *arXiv preprint arXiv:2304.02868*, 2023.

[125] Xu, Z., C. Yu, F. Fang, et al. Language agents with reinforcement learning for strategic play in the werewolf game. *arXiv preprint arXiv:2310.18940*, 2023.

[126] Light, J., M. Cai, S. Shen, et al. Avalonbench: Evaluating llms playing the game of avalon. *arXiv e-prints*, pages arXiv–2310, 2023.

[127] Shi, Z., M. Fang, S. Zheng, et al. Cooperation on the fly: Exploring language agents for ad hoc teamwork in the avalon game. *arXiv preprint arXiv:2312.17515*, 2023.

[128] Jin, M., B. Wang, Z. Xue, et al. What if llms have different world views: Simulating alien civilizations with llm-based agents. *arXiv preprint arXiv:2402.13184*, 2024.

[129] Liang, T., Z. He, J.-t. Huang, et al. Leveraging word guessing games to assess the intelligence of large language models. *arXiv preprint arXiv:2310.20499*, 2023.

[130] Qiao, D., C. Wu, Y. Liang, et al. Gameeval: Evaluating llms on conversational games. *arXiv preprint arXiv:2308.10032*, 2023.

[131] Shao, X., W. Jiang, F. Zuo, et al. Swarmbrain: Embodied agent for real-time strategy game starcraft ii via large language models. *arXiv preprint arXiv:2401.17749*, 2024.

[132] Li, K., A. K. Hopkins, D. Bau, et al. Emergent world representations: Exploring a sequence model trained on a synthetic task. *arXiv preprint arXiv:2210.13382*, 2022.

[133] Gupta, A. Are chatgpt and gpt-4 good poker players?–a pre-flop analysis. *arXiv preprint arXiv:2308.12466*, 2023.

[134] Wang, Z., Y. Y. Chiu, Y. C. Chiu. Humanoid agents: Platform for simulating human-like generative agents. *arXiv preprint arXiv:2310.05418*, 2023.

[135] Lin, J., H. Zhao, A. Zhang, et al. Agentsims: An open-source sandbox for large language model evaluation. *arXiv preprint arXiv:2308.04026*, 2023.

[136] Stengel-Eskin, E., A. Prasad, M. Bansal. Regal: Refactoring programs to discover generalizable abstractions. *arXiv preprint arXiv:2401.16467*, 2024.

[137] Côté, M.-A., A. Kádár, X. Yuan, et al. Textworld: A learning environment for text-based games. In *Computer Games: 7th Workshop, CGW 2018, Held in Conjunction with the 27th International Conference on Artificial Intelligence, IJCAI 2018, Stockholm, Sweden, July 13, 2018, Revised Selected Papers 7*, pages 41–75. Springer, 2019.

[138] Yuan, X., M.-A. Côté, A. Sordoni, et al. Counting to explore and generalize in text-based games. *arXiv preprint arXiv:1806.11525*, 2018.

[139] BBC. The hitchhiker's guide to the galaxy text adventure: 30th anniversary edition. https://www.bbc.co.uk/programmes/articles/1g84m0sXpnNCv84GpN2PLZG/the-game-30th-anniversary-edition.

[140] Shridhar, M., J. Thomason, D. Gordon, et al. Alfred: A benchmark for interpreting grounded instructions for everyday tasks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10740–10749. 2020.

[141] Chevalier-Boisvert, M., D. Bahdanau, S. Lahlou, et al. Babyai: A platform to study the sample efficiency of grounded language learning. *arXiv preprint arXiv:1810.08272*, 2018.

[142] Singh, I., G. Singh, A. Modi. Pre-trained language models as prior knowledge for playing text-based games. *arXiv preprint arXiv:2107.08408*, 2021.

[143] Sudhakar, A. V., P. Parthasarathi, J. Rajendran, et al. Language model-in-the-loop: Data optimal approach to learn-to-recommend actions in text games. *arXiv preprint arXiv:2311.07687*, 2023.

[144] Shi, Z., Y. Xu, M. Fang, et al. Self-imitation learning for action generation in text-based games. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 703–726. 2023.

[145] Ammanabrolu, P., M. Hausknecht. Graph constrained reinforcement learning for natural language action spaces. *arXiv preprint arXiv:2001.08837*, 2020.

[146] Guo, X., M. Yu, Y. Gao, et al. Interactive fiction game playing as multi-paragraph reading comprehension with reinforcement learning. *arXiv preprint arXiv:2010.02386*, 2020.

[147] Lewis, M., Y. Liu, N. Goyal, et al. Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. *arXiv preprint arXiv:1910.13461*, 2019.

[148] Wu, D., H. Shi, Z. Sun, et al. Deciphering digital detectives: Understanding llm behaviors and capabilities in multi-agent mystery games. *arXiv preprint arXiv:2312.00746*, 2023.

[149] Wikipedia. Portable game notation. https://en.wikipedia.org/wiki/Portable_Game_Notation, 2023.

[150] Toshniwal, S., S. Wiseman, K. Livescu, et al. Chess as a testbed for language model state tracking. In *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, pages 11385–11393. 2022.

[151] Gupta, A. Are chatgpt and gpt-4 good poker players?–a pre-flop analysis. *arXiv preprint arXiv:2308.12466*, 2023.

[152] Zhang, S., S. Roller, N. Goyal, et al. Opt: Open pre-trained transformer language models. *arXiv preprint arXiv:2205.01068*, 2022.

[153] Zhao, E., R. Yan, J. Li, et al. Alphaholdem: High-performance artificial intelligence for heads-up no-limit poker via end-to-end reinforcement learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, pages 4689–4697. 2022.

[154] Wu, S. A., R. E. Wang, J. A. Evans, et al. Too many cooks: Bayesian inference for coordinating multi-agent collaboration. *Topics in Cognitive Science*, 13(2):414–432, 2021.

[155] Puig, X., T. Shu, S. Li, et al. Watch-and-help: A challenge for social perception and human-ai collaboration. *arXiv preprint arXiv:2010.09890*, 2020.

[156] Gan, C., S. Zhou, J. Schwartz, et al. The threedworld transport challenge: A visually guided task-and-motion planning benchmark for physically realistic embodied ai. *arXiv preprint arXiv:2103.14025*, 2021.

[157] Gan, C., J. Schwartz, S. Alter, et al. Threedworld: A platform for interactive multi-modal physical simulation. *arXiv preprint arXiv:2007.04954*, 2020.

[158] Zhao, R., J. Song, Y. Yuan, et al. Maximum entropy population-based training for zero-shot human-ai coordination. In *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, pages 6145–6153. 2023.

[159] Significant Gravitas. AutoGPT.

[160] Kolve, E., R. Mottaghi, W. Han, et al. Ai2-thor: An interactive 3d environment for visual ai. *arXiv preprint arXiv:1712.05474*, 2017.

[161] Xia, F., W. B. Shen, C. Li, et al. Interactive gibson benchmark: A benchmark for interactive navigation in cluttered environments. *IEEE Robotics and Automation Letters*, 5(2):713–720, 2020.

[162] Savva, M., A. Kadian, O. Maksymets, et al. Habitat: A platform for embodied ai research. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 9339–9347. 2019.

[163] Srivastava, S., C. Li, M. Lingelbach, et al. Behavior: Benchmark for everyday household activities in virtual, interactive, and ecological environments. In *Conference on robot learning*, pages 477–490. PMLR, 2022.

[164] Li, C., R. Zhang, J. Wong, et al. Behavior-1k: A benchmark for embodied ai with 1,000 everyday activities and realistic simulation. In *Conference on Robot Learning*, pages 80–93. PMLR, 2023.

[165] Florensa, C., D. Held, X. Geng, et al. Automatic goal generation for reinforcement learning agents. In *International conference on machine learning*, pages 1515–1528. PMLR, 2018.

[166] Wang, Y., Y. Kordi, S. Mishra, et al. Self-instruct: Aligning language model with self generated instructions. *arXiv preprint arXiv:2212.10560*, 2022.

[167] Hao, S., T. Liu, Z. Wang, et al. Toolkengpt: Augmenting frozen language models with massive tools via tool embeddings. *Advances in neural information processing systems*, 36, 2024.

[168] Bender, E. M., A. Koller. Climbing towards nlu: On meaning, form, and understanding in the age of data. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 5185–5198. 2020.

[169] Bommasani, R., D. A. Hudson, E. Adeli, et al. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021.

[170] Li, Y., Y. Zhang, L. Sun. Metaagents: Simulating interactions of human behaviors for llm-based task-oriented coordination via collaborative generative agents. *arXiv preprint arXiv:2310.06500*, 2023.